photon's quantum wave had better split and travel both paths, so that an amalgam of the two can produce the observed interference pattern. But if the new detector turns out to have been on—or if it was switched on after the photon fully cleared the splitter—it would seem to present the photon with an identity crisis: on passing through the splitter, it had already committed itself to its wavelike character by traveling both paths, but now, sometime after making this choice, it "realizes" that it needs to come down squarely on the side of being a particle that travels one and only one path.

Somehow, though, the photons always get it right. Whenever the detector is on—again, even if the choice to turn it on is delayed until long after a given photon has passed through the beam splitter—the photon acts fully like a particle. It is found to be on one and only one route to the screen (if we were to put photon detectors way downstream along both routes, each photon emitted by the laser would be detected by one or the other detector, never both); the resulting data show no interference pattern. Whenever the new detector is off—again, even if this decision is made after each photon has passed the splitter—the photons act fully like a wave, yielding the famous interference pattern showing that they've traveled both paths. It's as if the photons adjust their behavior in the past according to the future choice of whether the new detector is switched on; it's as though the photons have a "premonition" of the experimental situation they will encounter farther downstream, and act accordingly. It's as if a consistent and definite history becomes manifest only after the future to which it leads has been fully settled.[4]

There is a similarity to your experience of deciding to sell *The Full Monty.* Before meeting with Mr. Smithers, you were in an ambiguous, undecided, fuzzy, mixed state of being both willing and unwilling to sell the painting. But talking together about the art world and learning of Smithers's affection for your great-uncle made you increasingly comfortable with the idea of selling. The conversation led to a firm decision, which in turn allowed a history of the decision to crystallize out of the previous uncertainty. In retrospect it felt as if the decision had really been made all along. But if you hadn't gotten on so well with Mr. Smithers, if he hadn't given you confidence that *The Full Monty* would be in trustworthy hands, you might very well have decided not to sell.